LinkForge Manual & Runbook

Master Guide: GitHub Synchronization, Antigravity Development, Fresh Windows Re-Installation, Extension Distribution & Cloudflare VPS Deployment

Local Project Path:	X:\.config\docker\linkforge
GitHub Repository:	https://github.com/siegewang/Linkforge.git
VPS & Cloudflare Domain:	https://linkforge.okapitek.uk (Tunnel → localhost:5006)
Database & Archives:	X:\.config\docker\linkforge\data\dashboard.db

1. Fresh Windows / VPS Setup & Clean Re-Installation

If you ever reinstall Windows or deploy on a Linux VPS behind Docker & Cloudflare Tunnel, follow these four simple steps:

Step 1: Install Prerequisites:

- **Docker & Docker Compose:** Install Docker Desktop (Windows) or docker-ce + docker-compose-plugin (Linux VPS).
- **Git:** Install Git from git-scm.com or apt install git.
- **Google Antigravity / Python 3.12:** Install Google Antigravity coding environment (for local development).

Step 2: Clone the Project from GitHub:

Open terminal / PowerShell and run:

```
git clone https://github.com/siegewang/Linkforge.git
cd Linkforge
```

Step 3: Launch the Container:

Start the containerized LinkForge instance (runs on port 5006:5000):

```
docker compose up -d --build
```

Step 4: Restore Data (Optional):

Open https://linkforge.okapitek.uk/admin/data (or http://localhost:5006/admin/data), click **Restore from Backup / CSV**, and select your latest linkforge_auto_backup_*.zip. All bookmarks, full texts, notes, and video categories will be populated automatically.

2. How to Make Changes & Develop with Antigravity

When you want to add features, modify styling, adjust AI ingestion, or fix anything in LinkForge with Antigravity:

Opening the Project in Antigravity:

1. Launch Google Antigravity.
2. Open the directory workspace: X:\.config\docker\linkforge.
3. Antigravity automatically detects all blueprints, services, templates, and extension folders.

Example Prompts to Give Antigravity:

- *'Add a new tag filter in the Neural Links sidebar that sorts tags by frequency.'*
- *'Update the YouTube transcript extractor to support additional subtitle languages.'*
- *'Change the theme accent colors in base.html and ensure mobile responsive padding is clean.'*
- *'Package the browser extensions and update both Chrome and Firefox ZIP bundles.'*

Codebase Structure for Reference:

Directory / File	Description & Responsibility
blueprints/	Flask route controllers (dashboard, bookmarks, videos, notes, admin, api)
services/	Core logic (backup.py, updater.py, nuke.py, db.py, scraper.py)
templates/	Jinja2 HTML frontend views (base.html, index.html, bookmarks.html, videos.html, admin_*.html)
dashforge-extension/	Browser extension source code (Manifest V3 for Chrome & Firefox)
data/	SQLite database (dashboard.db), backups/, and offline archives/ (Ignored by Git)
docker-compose.yml	Container definition, port mapping (5006:5000), and volume bindings

3. How to Update Your GitHub Repository

Whenever you or Antigravity complete code changes, push them to GitHub with this standard 3-command workflow:

```
# 1. Navigate to the project folder
cd X:\.config\docker\linkforge
# 2. Stage all modified files and commit with a description
git add .
git commit -m "Describe what you changed here (e.g. Add dark mode toggle)"
# 3. Push to your GitHub repository
git push origin main
```

100% Privacy Guarantee: The .gitignore configuration strictly excludes your ./data/ folder, SQLite databases (*.db), cached article archives, and private secrets. Pushing to GitHub will NEVER upload your bookmarks, personal notes, or login credentials.

4. Remote VPS Deployment, Cloudflare Tunnel & Live Updates

How to deploy on your VPS behind a Cloudflare Tunnel and receive instant live updates:

Cloudflare Tunnel Configuration (linkforge.okapitek.uk):

In your Cloudflare Zero Trust / Tunnel dashboard, add a Public Hostname rule:

• **Subdomain:** linkforge • **Domain:** okapitek.uk

• **Type:** HTTP • **URL:** localhost:5006 (or linkforge:5000 if sharing Docker bridge)

• *Note: LinkForge includes ProxyFix middleware, ensuring HTTPS redirects, PWA Web Share Target, and client IPs operate with zero configuration over Cloudflare.*

How Future Updates Work (Zero-Terminal / 100% In-Browser):

1. You push new code to GitHub from your PC (git push origin main).
2. Within seconds, your VPS and remote instances detect the new GitHub commit.
3. A glowing green/cyan badge ■ **UPDATE AVAILABLE** appears in the left sidebar.
4. In Settings, click **'Update & Restart Now'** — LinkForge pulls the update and restarts in 2 seconds!

Cleaning Test Data (The Nuke Data Tab):

To clean restored test data on a fresh VPS while keeping all AI configuration:

1. Open Admin Panel ➔ Nuke Data (/admin/nuke).
2. Click **'Preset: Clean Restored Data (Keep AI Settings)'**.
3. Type **NUKE** in the confirmation box and click **Confirm Nuke**.

5. Troubleshooting & Maintenance Commands Cheat Sheet

Action / Problem	Command to Run in PowerShell / Bash
Restart LinkForge Container	`docker compose restart`
Rebuild from scratch (forced)	`docker compose up -d --build --force-recreate`
View live application logs	`docker logs -f linkforge`
Discard local unpushed changes	`git reset --hard origin/main`
Repackage browser extensions	`python package_extensions.py`

6. Firefox Extension (.XPI) Signing & In-App Distribution

Mozilla Firefox enforces strict cryptographic signatures on all add-ons. To distribute your LinkForge extension to users on standard release Firefox with **zero security warnings and 1-click installation**, sign the add-on through Mozilla's free self-distribution portal and save it to the LinkForge static downloads directory.

Method 1 — Official Mozilla Web Signing & LinkForge In-App Serving (Recommended)

Follow these exact steps to get your extension signed and hosted directly inside LinkForge:

Step 1: Access Mozilla Developer Portal
Go to **https://addons.mozilla.org/developers/addon/submit/distribution** and log in with your Mozilla account.

Step 2: Select Self-Distribution
Choose **'On your own'** (this signs the file for direct distribution without publishing to the public AMO store).

Step 3: Upload the Packaged Zip File
Click *'Select a file...'* and upload:
X:\.config\docker\linkforge\static\downloads\dashforge-firefox.zip

Step 4: Download the Signed XPI
Mozilla's validator will verify the manifest (passes with 0 errors) and sign the package in ~1-2 minutes. Download the resulting signed .xpi file.

Step 5: Place the Signed XPI in LinkForge Static Downloads (CRITICAL)
Rename / save the signed file to replace the existing file at:
X:\.config\docker\linkforge\static\downloads\dashforge-firefox.xpi
(Path inside Docker container: /app/static/downloads/dashforge-firefox.xpi)

Step 6: In-App 1-Click Installation on Remote Computers
Now, whenever anyone visits LinkForge at **Admin → Extensions** (https://linkforge.okapitek.uk/admin/extensions) and clicks **'Download for Firefox'**, they download this officially signed .xpi. Firefox will install it permanently with a single click and **zero warning screens**!

Alternative CLI Signing with web-ext:

You can also automate signing via terminal: web-ext sign --api-key="..." --api-secret="..." --source-dir=dashforge-extension --artifacts-dir=static/downloads

Extension Settings on Remote Instances: When opening the extension popup on family/remote computers, set the server URL to https://linkforge.okapitek.uk to log articles and videos directly to your cloud VPS.